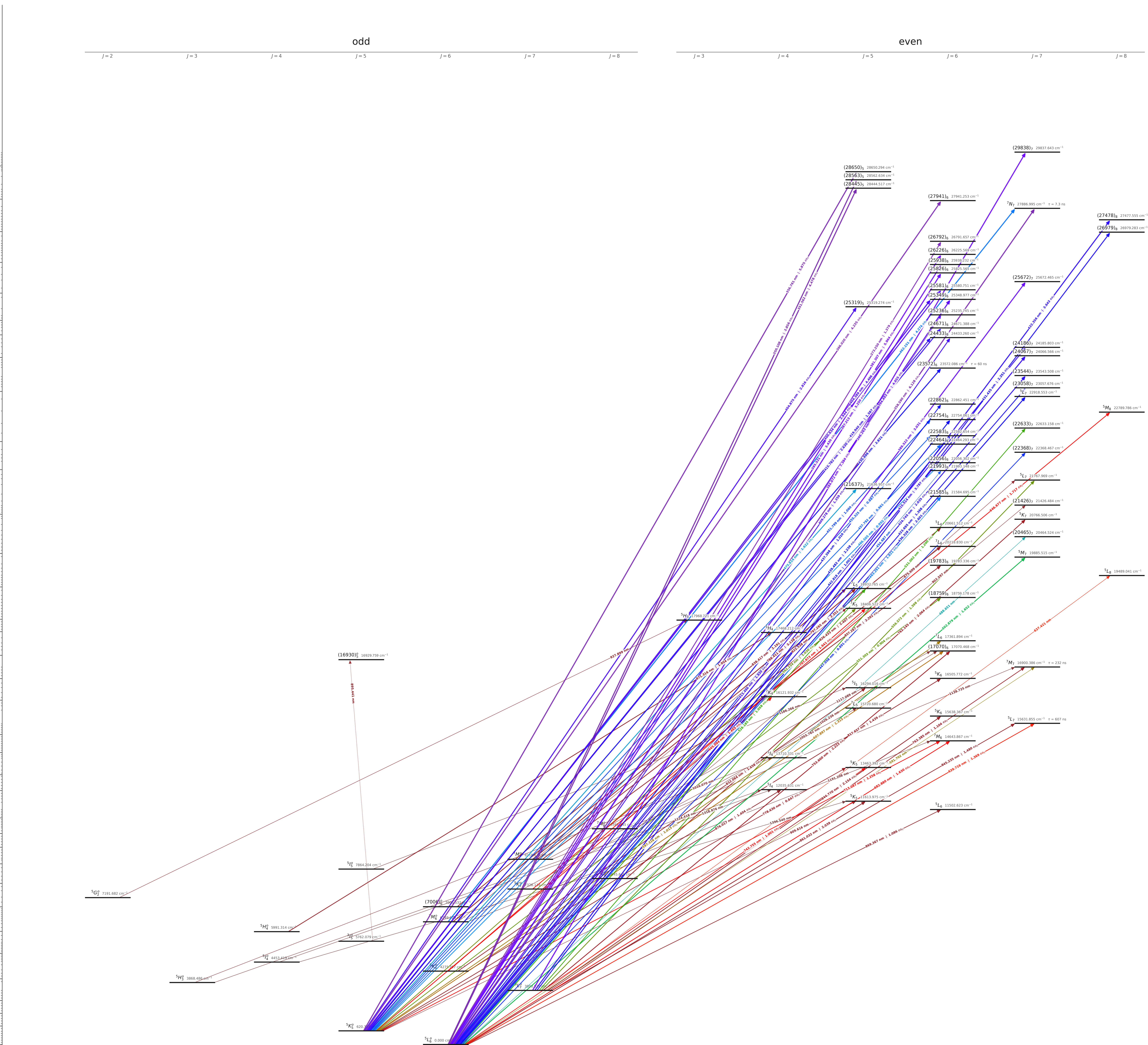


U⁺ ionisation limit 49958.40 cm⁻¹ = 6.19405 eV (200.167 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle V \rangle_{er} / \hbar|$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ [J I'] $ (ea)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$^2L_{3/2} - (^2S_{1/2})_0$	396.2633	396.1512	756.5486	-	-	-	-	396.263 nm
$^2L_{3/2} - ^2M_2$	591.7025	591.5385	506.6068	-	-	0.686	-	591.54 nm: classic first-step line of uranium laser spec...
$^2L_{3/2} - ^2L_1$	639.7193	639.5425	428.6312	1.39	9.670e+05	0.2622	58.6	639.542 nm
$^4K_0 - ^2L_1$	713.0805	712.8900	420.4153	1.258	6.800e+05	-	-	713.087 nm
$^2L_1 - ^2M_1$	763.3846	763.1745	292.7149	1.104	3.700e+05	0.686	8.58	763.385 nm
$^7M_0 - ^2L_1$	788.4112	788.1904	380.2488	2.138	3.700e+06	-	-	788.411 nm: line used for Doppler-free saturated absorpt...
$^2L_1 - ^4K_0$	861.0316	860.7952	348.1782	3.109	2.200e+06	-	-	strongest ground-state absorption

λ , ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

N ₂ hyperfine structure (J = 7/2)			
Level	A (MHz)	B (MHz)	Shift of each level from the centre of gravity (MHz)
¹ _g	-60.540	404.010	F=5/2: +3103.57 F=7/2: +2197.25 F=9/2: +1190.70 F=11/2: +202.90 F=13/2: -620.41 F=15/2: -1107.31 F=17/2: -1059.96 F=19/2: -258.84
³ _g	-68.346	417.054	F=5/2: +1452.64 F=7/2: +1275.57 F=9/2: +1020.91 F=11/2: +714.29 F=13/2: +433.13 F=15/2: -112.36 F=17/2: -619.70 F=19/2: -112.02
⁵ _g	-56.310	412.000	F=7/2: +3113.03 F=9/2: +2118.90 F=11/2: +1081.36 F=13/2: +106.94 F=15/2: -678.50 F=17/2: -1129.70 F=19/2: -1082.05 F=21/2: -351.90
⁷ _g	-59.130	447.000	F=5/2: +1626.47 F=7/2: +1343.32 F=9/2: +966.69 F=11/2: +599.63 F=13/2: +168.12 F=15/2: -278.98 F=17/2: -719.91 F=19/2: -129.68
⁹ _g	-67.890	2366.000	F=5/2: -72.08 F=7/2: -886.07 F=9/2: -1007.31 F=11/2: -1014.66 F=13/2: -823.62 F=15/2: -334.33 F=17/2: -568.45 F=19/2: +2015.30
¹¹ _g	54.300	609.000	F=5/2: +1574.77 F=7/2: +1280.60 F=9/2: +926.11 F=11/2: +529.09 F=13/2: +111.29 F=15/2: -301.58 F=17/2: -479.86 F=19/2: -989.94
¹³ _g	-69.000	609.000	F=9/2: -172.75 F=11/2: +115.12 F=13/2: +563.33 F=15/2: -19.71 F=17/2: -540.22 F=19/2: -972.60 F=21/2: -1237.42
¹⁵ _g	-99.610	5962.042	F=7/2: +2484.44 F=9/2: +2090.09 F=11/2: +1569.85 F=13/2: -955.57 F=15/2: -1288.20 F=17/2: -382.21 F=19/2: -296.51 F=21/2: -1486.45
¹⁷ _g	20.301	4950.120	F=5/2: +1476.18 F=7/2: +702.83 F=9/2: -98.63 F=11/2: -783.57 F=13/2: -1175.21 F=15/2: -1064.63 F=17/2: -079.19 F=19/2: +1659.53
¹⁹ _g	-84.720	2425.000	F=7/2: +4024.34 F=9/2: +2846.51 F=11/2: +1597.85 F=13/2: +339.82 F=15/2: -632.99 F=17/2: -1323.61 F=19/2: -1501.94 F=21/2: -970.94
²¹ _g	-171.000	2685.000	F=7/2: +5791.19 F=9/2: +4538.01 F=11/2: +3122.26 F=13/2: +1613.49 F=15/2: -93.99 F=17/2: -1341.70 F=19/2: -2585.79 F=21/2: -3518.25

Measured A, B constants; sources in the data file

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^4\text{Li}^-$ (28563),	350.108	235-238	10340.0
$^4\text{Li}^-$ (288445),	351.562	235-238	8990.0
$^4\text{Li}^- - \text{H}$,	358.590	235-238	17480.0
$^6\text{Li}^-$ (27941),	366.020	235-238	4800.0
$^6\text{Li}^-$ (26792),	373.250	235-238	5910.0
$^6\text{Li}^-$ (26626),	381.307	235-238	6420.0
$^6\text{Li}^-$ (29838),	384.072	235-238	13800.0
$^6\text{Li}^-$ (29854),	385.552	235-238	8950.0
$^6\text{Li}^-$ (25826),	387.213	235-238	8180.0
$^6\text{Li}^-$ (25672),	389.522	235-238	10880.0
$^6\text{Li}^-$ (25349),	394.493	235-238	8420.0
$^6\text{Li}^-$ (25236),	396.263	235-238	12600.0
$^6\text{Li}^-$ (25581),	400.634	235-238	4320.0
$^6\text{Li}^-$ (25349),	404.389	235-238	8570.0
$^6\text{Li}^-$ (25319),	404.875	235-238	13280.0
$^6\text{Li}^-$ (24433),	405.578	235-238	5630.0
$^6\text{Li}^-$ (24433),	409.214	235-238	10500.0
$^6\text{Li}^-$ (24067),	419.782	235-238	12700.0
$^6\text{Li}^-$ (24671),	415.782	235-238	11390.0
$^6\text{Li}^-$ (24433),	419.940	235-238	5820.0
$^6\text{Li}^-$ (27478),	422.356	235-238	10460.0
$^6\text{Li}^-$ (23544),	427.435	235-238	8210.0

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$